A Shorthand Tale about Life, Health and Longevity
(for beginners and medical doctors)

Most medical doctors (MDs) exercise their profession without being really familiar with the process of biological evolution that generated human organism and previously all other life forms on Earth. This process goes on for about 4 billion years, proceeding in countless successive steps to construct ever more complex organisms. Hence, how our organism works can be understood only in the light of evolution (Th. Dobzhansky's famous saying). MDs, as professionals of maintenance of human organism, mainly the healing its diseases, by repairing diverse biological dysfunctions, should be familiar with the process that made human organism. I do not want to offend either the minority of MDs who are familiar with the science of evolution or the majority who are not. I would not mind offending those MDs who are arrogant while ignoring human evolutionary biology while applying routine medical protocols.

Traditionally, law and medical schools cultivate defensive arrogance whenever their work is considered questionable. The impact of these two professions on people's destinies is huge: law can sentence anybody, even by mistake or influence, to prison or death, and the medical practice can be decisive about health and disease, survival and death. Monumental neo-colonial edifices of law and medical schools signal the societal impact of these two professions. Law and medicine bestow power and power does not inspire modesty and learning (they have already learned, that's why they have power!). In addition, power acts as an elixir of life in displaying the sex appeal of confidence, energy and health. It's all about the same thing: enjoying life *versus* fearing death! That's why research on biology of aging and deadly diseases gets the attention of the public.

But, as soon as cell biology, and its research funding, got close to medicine and pharma companies, cell biologists did not resist the sex appeal of power and the quality of their research readily dropped. The effect can be seen in intellectual impoverishment of mainstream biomedical research and frequent lack of reproducibility of published results from academia. Academic culture got contaminated by corporate culture and the beauty of

molecular biology was replaced by the dullness of cell biology and cell biologists eager to please the taste of pharma industry to get closer to their money. I would not publicly present this personal point of view, which will not please the biomedical establishment (including governmental funding agencies), if it were not shared - but only in private - by vast majority of leading biologists whom I have interviewed. Scientists are most fearful about their chances to get the money (grants) for their research. That money being in the hands of powerful establishments, scientists avoid offending such establishments.

What is life?

Back to fundamentals: what is life (in the light of evolution)? This question is the title of a brilliant small book published in 1944 by the famous physicist Erwin Schroedinger. He wrote that, basically, life is a complex structure composed of cells filled with thousands of diverse kinds of molecules that cooperate together in harmony (or homeostasis) to create, maintain and grow a functioning order. The rest of the expanding universe undergoes increasing disorder, or entropy. Living organisms, even the microorganisms, are tiny ephemeral islands of order, i.e., islands of negative entropy (called negentropy by Schroedinger), in the ocean of growing entropy. Schroedinger contends that to create and maintain order, energy must be constantly pumped into the system. Life is a metastable state whose stable state is to be dead, wrote Schroedinger. Only sufficient energy allows a living organism to last. The energy is stored and incorporated into the structures of atoms and molecules and can be liberated by decomposing such structures. For instance, cells get their energy by decomposing relatively complex molecules into smaller and simpler ones and high-jacking the liberated energy to build their own structure. Thus, life “feeds” on order of molecular structures by their decomposition. Life destroys microscopic order to get the energy to create macroscopic order. Cells grow by producing excess of complex molecular structures that results in cell’s growth. When the cell doubles its size, it divides into two nearly identical “daughter cells” and so on. That’s the very beginning of the tale of life for beginners, before even mentioning DNA and proteins.

Thus, building and maintaining the order of a living organism requires predation upon complex structures and release of simpler structures. For instance, in our organism, sugar and fat molecules are decomposed - “burned” - by consuming oxygen and releasing and evacuating water and carbon dioxide by urinating and exhaling, respectively.

Interestingly, life itself produces sugars as the main source of energy required for living and growth. How is the energy stored in sugars during their biosynthesis? By a biological process called photosynthesis. Within particular bacteria and plankton, and in all plants, a pigment called chlorophyll is used to absorb solar light energy to compose sugar (glucose) molecules from water and carbon dioxide. It's just like Schroedinger said: energy is used to make higher order (a sugar molecule) by fusing lower order structures (water and carbon dioxide). Hence, first, the original solar light energy is stored in the chemical bonds between the atoms composing the glucose structure. Then, glucose, and some other complex molecules like fat and proteins, are used as batteries liberating energy by their decomposition and investing that energy into chemical bonds to synthesize all biological molecules such as proteins, DNA, lipids and other. The energetic intermediate and universal transmitter and deliverer of biological energy is a small energy-rich molecule called ATP.
The energy for constructing atoms from elementary particles is so high that decomposing atoms liberates "atomic energy" displayed by the explosion of atomic bomb or by nuclear power plants. The energy investment to make higher order structures goes up to societal structures like the European Union. Can somebody calculate the energy cost to make United States of America or the European Union?

The Story of Life and its Maintenance

But, let's first consider the story of life which is the story of integrations of smaller into larger structures. Estimates are that life on Earth started about 4 billion years ago in the form of unicellular, bacterium-like, anaerobic organisms - probably sluggish and inefficient due to low availability of energy (ATP). With low atmospheric oxygen, there was very little to no ozone, therefore the solar ultraviolet (UV) light was so intense that it would kill all cells exposed to UV. Life could persist only while protected from UV by water layer.

Hence, early life was aquatic, anaerobic and unicellular until the Great Oxygenation Event occurred some 3bn years ago. By that time, cyanobacteria acquired photosynthesis using solar energy, consuming carbon dioxide and water to make sugar and release oxygen. Eventually, atmospheric oxygen concentration rose from about 1% to final 21% - sufficient to build ozone (fused molecular oxygen) layer around the Earth that absorbed the majority of

damaging UV light. Only then, it became safe for life to grow on water surface and outside of the seas and lakes.
Sugar molecules made by photosynthesis cannot by themselves drive cell's biochemistry, but ATP can. Eventually, oxidative metabolism emerged that could use sugar to make ATP, with oxygen as the catalyst to "burn" sugar. This process, called oxidative phosphorylation, produced a lot of ATP (good news) but generated also a lot of chemically aggressive toxic side product - oxygen free radicals (or ROS) - which was bad news for life. ROS produced by energy metabolism are of similar kind to ROS generated by UV light, ionizing radiation and many toxic chemicals. Experts believe that during the Great Oxygenation Event life almost got extinct due to the corrosive effect of ROS on living cells. We can now observe such kind of oxidative disaster by exposing today's strictly anaerobic bacteria to the atmospheric oxygen: no survivors are left. ROS is best antibiotic for anaerobic organisms, therefore it is used by our immune system to kill bacteria. Evolution keeps trying and eventually finds the use for even for most abhorred things. Obviously, life did not get extinct because of ROS.

Proteins hold the key to life, DNA holds the key to proteins

We cannot know well the details of so distant past, but we know that proteins from today's aerobic (oxygen consuming) organisms, including us, are remarkably resistant to ROS due to evolutionary selection of protein structures resistant to corrosion by ROS. Most of our proteins are quasi inoxidable ("inox", like the materials from expensive cars and solid kitchen appliances)! The market value of materials is proportional to their resistance to oxidation, from precious diamonds to banal iron. The same criterion applies to biological "market value": the lower the level of protein damage by oxidation, the longer lived is the organism. We have a pretty good idea how proteins resist oxidation via special structural features.
If humans and other aerobic creatures possess "inox" proteins, why don't we live longer? Because we already "live longer" - as long as our cells can keep protein corrosion at low level - and because the oxidation-resistance feature of proteins is itself fragile. Subtle deviations from the best "native" structure (e.g., due to random mistakes in protein production or to fixed errors called "silent mutations" or polymorphisms) sensitize the affected protein to oxidation. The fragile "polymorphic" proteins can originate in DNA sequence and remain "silent" (functionally inconsequential) for most of the lifespan. However, when a critical level

of proteins damaged by oxidation is reached, the functional “silence” is broken by protein dysfunction and onset of disease. Such “silent” mutation is the physical basis of (at least a fraction of) innate predispositions to age-related disease: they are our “weak links”.

Why do we talk so much about proteins and so little about DNA (genes)? Because all life functions are carried out by proteins - either directly or via molecules made by proteins. The research on special bacteria and small animals that resist extremely high radiation doses showed that their resistance is due to protection and defence of proteins - not DNA - against oxidation by ROS which are produced in cells by radiation. Intact proteins specialized in repair of DNA damage will fix even a massacred, highly radiation-damaged, DNA. Cells die when their damaged proteins cannot properly do their work, which includes making ATP (energy) and repairing, replicating and reading DNA to make more of new proteins that do everything. Cell properties (phenotype) change because protein activities are changed. Only when DNA mutation changes the amount or quality of protein it will change the phenotype of cells and organism. Some very rare mutations produce “desirable” effects by changing protein activity and improving the performance of cells and organisms. Dysfunction of protein, or proteins, will cause undesirable deleterious phenotypes. A class of undesirable phenotypes are called diseases!

Why is Aging linked to Diseases?

Which are the age-related diseases? All diseases that increase in human population exponentially with person’s age (about 5th power of time). They are: rheumatoid arthritis and cardiovascular, malignant, neurodegenerative and immunity-related diseases and conditions that account for about 90% of mortality in economically developed countries (and progressively so in other countries). This is why such diseases are of utmost importance.

By identifying and quantifying oxidative damage to each individual protein species, our research opened the path to the development of predictive diagnostic, i.e., identification of the inborn predispositions to diseases, at any point in life. Once we know the predisposition, can we do something about the predicted forthcoming disease? Yes, but this is now a matter of research and development to start nothing less than interfering with bad luck or destiny.

A disease genetically programed to occur at some age will not happen because we would know how to prevent its onset!
Particular natural molecules, so far all bright-coloured pigments isolated from extremely robust (radiation-resistant) bacteria can act as so-called chemical chaperones. Such chemical chaperones bind preferentially to oxidation-sensitive proteins and protect them from oxidation. Chemical chaperones appear as the first-in-line candidates for mitigation of both sporadic and genetically predisposed age-related diseases. This means that soon we might be able - for the first time in human history - to intervene into our inborn health-related destiny. Instead of falling victim to disease pre-programmed by genetic predisposition, one can expect to keep the potential disease attenuated, in long dormancy, for an indeterminate time. One can treasonably expect even to reverse the course of an already manifested disease!

This part of the story may sound as wishful thinking or exaggerated optimism by drawing a too simplified picture? What about other star candidates for the cause of senescence, ageing and ARD (e.g., telomeres, metabolism, epigenetic alterations, protein misfolding and aggregation, etc.)? Are they at once irrelevant or wrong? No, to the contrary, but they are all the consequences of age/oxidation-related dysfunction of relevant proteins (e.g., telomerase, DNA methylases, chaperones, ROS-detoxifying enzymes, etc.) which is the core concept promoted elsewhere* and here. The well-known proteins and processes are not alternatives to this story since they are included, interpreted and integrated by the new concept. Their effects can be deduced from this concept by invoking dysfunction of relevant proteins, or the acquisition of toxic functions by the accumulation of oxidatively damaged proteins (ref). The new concepts and the supporting knowledge are either wrong or right. Since we could not prove them wrong, they are likely to be right, and, if this were the case, slowing down further research and development becomes unethical.

About Selection and Adaptation

To us, Mother Nature is not a tender parent. When we admire the perfection of its living products - be it flowers, butterflies or ourselves - we should remember that we are looking only at living creatures and recall that the majority of them die early death. In 19th century, half of healthy born children died before the age of ten! Death was a cruel punishment of

less fit infants - called natural selection - a key term in the theory of evolution. Today, in rich countries, babies and children die rarely to the delight of their parents, but this means that there is little to no selection such that the “perfection” of Mother Nature will become less impressive. High survival will thwart natural selection and the accumulation of genetic defects (that do not kill us before the reproductive age) will necessarily follow. Cultural evolution, i.e., knowledge, may support the survival of genetically unfit individuals and create the dependence of a growing fraction of human population upon diverse kinds of prostheses, from mechanic and electronic to chemical. Such dependence will bring about new trans-humanistic industries, especially electronic and pharmaceutical, and novel societal controls upon individuals. Our descendants may work for the profit and power of insurance companies.

Resetting of the Society

Until now, Western society considered that we are fit enough to work to the retirement age of about 65 years. In the not so distant future, the cost of human biological maintenance may rise to untenable levels added upon accumulation of other societal problems. Too many unsolved “small problems” can become a huge global problem handicapping or paralysing the society to its dysfunctionality. Already now, the fear of unsolvable problems and of general uncertainty brings about growing public unrest. Fixing growing numbers of limited problems becomes inefficient and does not look as sustainable. Rather, a resetting of the whole society seems unavoidable, like when small operational mistakes cause a computer to idle. When this happens, we turn the computer off and on to reinitiate or reset the programs back to functionality. Sometimes, this move does not suffice and we need to reinstall old programs or look for better programs to replace the obsolete ones.

Traditionally, societal resetting occurred after dramatic wars with massive deaths, psychological traumas and infrastructural devastation. Wars sweep away the junk of numerous accumulated problems and people concentrate on essentials - on life itself. Common sense is back, productivity is back, and the society takes off with new optimism.

Is war the only option to reset the society? Hopefully not. Creating miracles is another option. The catholic church used them when needed. Miracles are major cultural (scientific, technological, artistic, etc.) transitions, like electricity (remember Tesla’s shows?), Sputnik (alone, with a dog, with a man…), television, computers, iPhones, etc., or some amazing

jokes… I don’t know. Miracle has to be a surprise, and one cannot plan a surprise (no milestones and deliverables for surprise!). Humor is great - you cannot have a good joke and lough if it’s not a surprise! Carlie Chaplin, Buster Keeton, Monty Pythons,. I created a small private institute which is “investment in surprise”

Even my name prevents me to mess with wars. I was born in Yugoslavia towards the end of WWII and my mother gave me the name which means “the one who celebrates peace”!

So, I am left with miracles. A miracle would be to live 200 years or just to get rid of the fear of dying…

Transhumanism and Biological Rejuvenation will compete…